IDENTIFYING THE MOST RELIABLE PLAYERS UNDER PRESSURE

SupeRR Selector Season 2 — Round 1 Technical & Analytical Methodology Report

Executive Summary & Challenge Statement: Delivering when the game is on the line is cricket's ultimate differentiator. Analyzing 200,999 delivery records across 867 IPL and T20I matches (2022–2026), we develop the **Pressure Reliability Index (PRI)**. We construct a continuous *Pressure Index (PI ∈ [0, 1])* for every ball, isolate high-pressure events (PI ≥ 0.60, the top 33.4% of crunch deliveries), and quantify each player's performance across four pillars: Absolute Execution (shrunk via Empirical Bayes), Clutch Differential vs Baseline, Contextual Value (Runs Above Expectation / Runs Saved), and Cross-Match Consistency. Below are the definitive Top 10 rankings for Unstop dropdown submission.

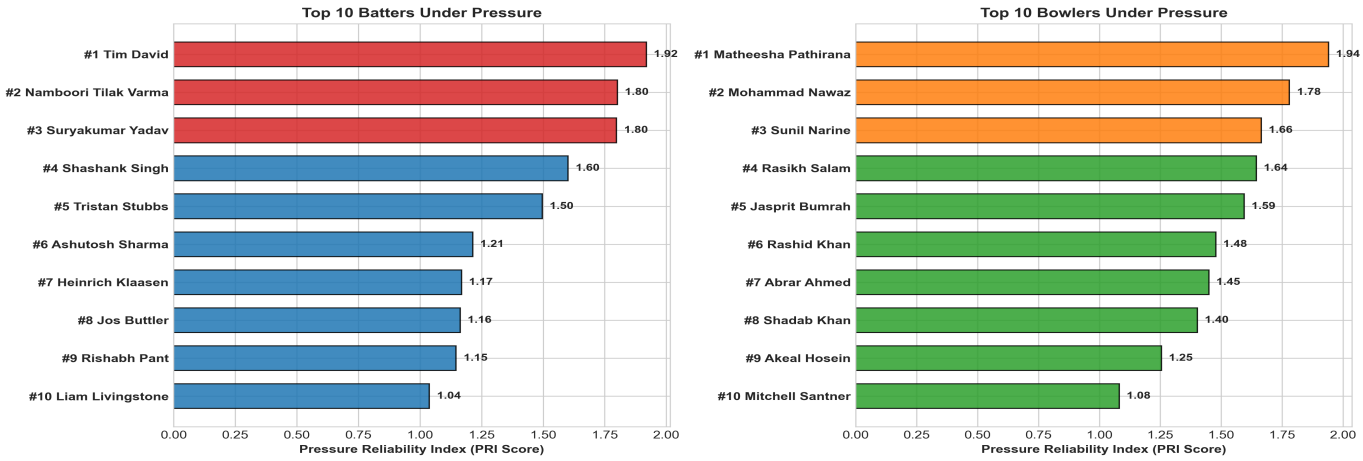
Table 1: Top 10 Batters Under Pressure (Ranked 1 to 10 for Dropdown Selection)

Rank	Player Name	HP Balls	HP Runs	HP SR	Base SR	Δ SR	HP Avg	RAA (Runs)	Consist. %	PRI Score
#1	**Tim David**	770	1445	187.7	160.7	**+24.4**	35.2	+403.2	88.4%	**1.92**
#2	**Namboori Tilak Varma**	962	1629	169.3	126.7	**+41.6**	45.2	+354.6	71.7%	**1.80**
#3	**Suryakumar Yadav**	832	1516	182.2	159.6	**+20.6**	35.3	+444.0	84.1%	**1.80**
#4	**Shashank Singh**	397	712	179.3	118.4	**+57.3**	44.5	+188.6	71.4%	**1.60**
#5	**Tristan Stubbs**	616	1058	171.8	121.0	**+48.9**	40.7	+230.3	75.9%	**1.50**
#6	**Ashutosh Sharma**	307	544	177.2	82.5	**+90.4**	27.2	+116.9	81.2%	**1.21**
#7	**Heinrich Klaasen**	704	1245	176.8	156.8	**+18.1**	31.9	+327.9	86.1%	**1.17**
#8	**Jos Buttler**	431	793	184.0	150.3	**+29.9**	31.7	+209.6	87.5%	**1.16**
#9	**Rishabh Pant**	410	720	175.6	127.0	**+45.5**	31.3	+171.4	87.0%	**1.15**
#10	**Liam Livingstone**	482	853	177.0	132.2	**+41.9**	25.8	+226.4	85.2%	**1.04**

Table 2: Top 10 Bowlers Under Pressure (Ranked 1 to 10 for Dropdown Selection)

Rank	Player Name	HP Balls	HP Wkts	HP Econ	Base Econ	Δ Econ	HP SR	Saved (Runs)	Consist. %	PRI Score
#1	**Matheesha Pathirana**	645	59	8.69	8.98	**+1.04**	10.9	+226.2	91.9%	**1.94**
#2	**Mohammad Nawaz**	214	20	5.80	7.53	**+1.76**	10.7	+75.6	81.8%	**1.78**
#3	**Sunil Narine**	441	31	6.46	7.14	**+1.13**	14.2	+167.2	90.0%	**1.66**
#4	**Rasikh Salam**	262	30	9.05	10.24	**+1.95**	8.7	+17.5	87.5%	**1.64**
#5	**Jasprit Bumrah**	743	50	7.22	7.34	**+0.76**	14.9	+290.0	81.8%	**1.59**
#6	**Rashid Khan**	920	77	7.80	7.36	**+0.27**	11.9	+128.0	87.2%	**1.48**
#7	**Abrar Ahmed**	256	19	6.09	7.09	**+1.18**	13.5	+90.3	92.9%	**1.45**
#8	**Shadab Khan**	238	20	7.13	8.22	**+1.45**	11.9	+45.1	90.9%	**1.40**
#9	**Akeal Hosein**	282	19	6.55	7.99	**+1.76**	14.8	+78.4	92.3%	**1.25**
#10	**Mitchell Santner**	480	32	7.34	7.51	**+0.75**	15.0	+115.6	91.7%	**1.08**

1.1 Visualizing the Top 10 Pressure Rankings (PRI Scores)



2. Pressure Definition: Situations, Indicators & Mathematical Architecture

Match pressure is fundamentally defined by **outcome leverage**: the degree to which a single delivery alters win probability. A sound definition must isolate high-stakes game states while filtering out ordinary consolidation phases and dead blowouts.

The Dual-Innings Pressure Model:

A. Innings 2 (The Chase): Pressure is dynamically calculated on every ball from four interacting mechanics:

• **Required Run Rate (RRR) Escalation:** Baseline T20 scoring rate is ~8.5 RPO. As RRR climbs from 9.5 to 18.0, pressure scales linearly: $p_{RRR} = clip((RRR - 5.0) / 9.0, 0.10, 1.00)$.

• **Resource Scarcity (Wickets in Hand):** Wickets in hand ($W_{rem} = 10 - wickets_lost$) dictate risk tolerance. Losing 7–8 wickets concentrates catastrophic risk on remaining batters ($p_w = (10 - W_{rem}) / 9.0$).

• **Time Scarcity / Stage Urgency:** As legal balls remaining deplete ($B_{rem} \leq 30$ in overs 16–20), every dot ball drastically decreases win probability: $p_{urgency} = 0.30 + 0.70 \times (120 - B_{rem}) / 120$.

• **Viability Filter (Excluding Blowouts):** Chases requiring > 4 runs/ball with > 6 balls left collapse into dead time (scaled down by 0.40). Live finishes (overs 17–20 with reachable equations) receive peak ratings (≥ 0.75 to 0.95).

B. Innings 1 (Target Setting & Crisis Recovery): In the first innings, pressure emerges from two structural phases:

• **Top-Order Collapse (Crisis Batting):** Losing ≥ 2 wickets inside the Powerplay (overs 1–6) or ≥ 4 wickets before over 12 triggers crisis pressure. Incoming batters face immense strain to rebuild without stalling team momentum.

• **Death Phase Execution (Overs 16–20):** In modern T20, death overs dictate whether a total is 155 (sub-par) or 195 (winning). Bowlers execute against aggressive slogging with boundary fielding restrictions; batters risk wickets seeking boundary acceleration.

C. Tournament Stakes & Super Overs:

• **Playoff & Knockout Multiplier:** Knockout matches (Final, Eliminator, Qualifier, Semi Final, Super 4) carry existential elimination risk and receive a **1.15× multiplier**.

• **Super Overs:** Deliveries in Super Overs (Innings ≥ 3) are assigned absolute maximum pressure ($PI = 1.00$).

2.1 Empirical Validation of Pressure Dynamics Across Match Phases

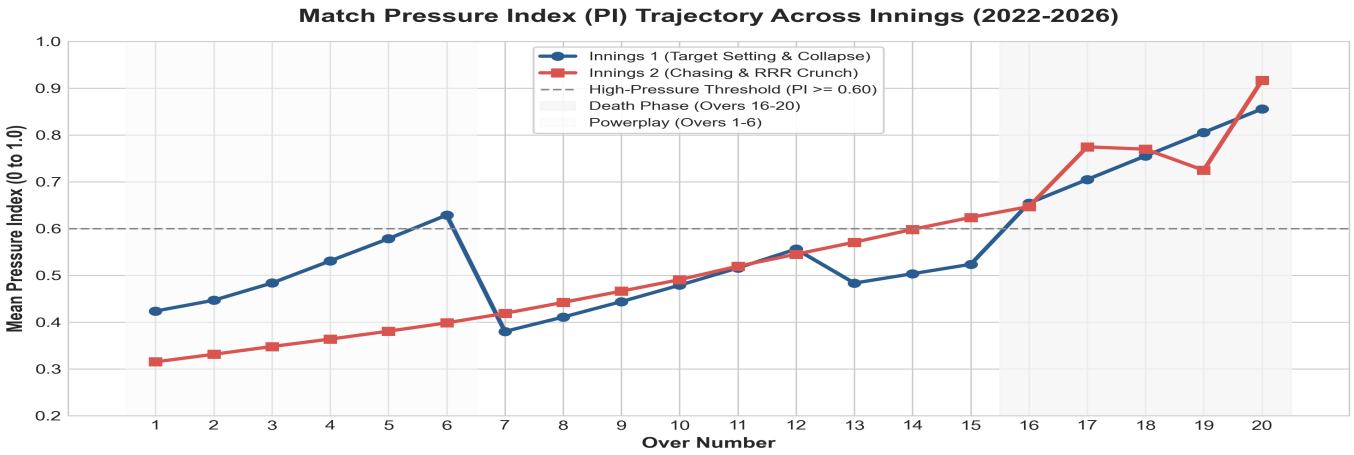


Figure 1: Mean Pressure Index across 20 overs. Note the sharp escalation in death overs (overs 16–20) and second-innings chases, peaking above 0.90 in final over finishes.

3. Performance Measurement: The Four Pillars of the Pressure Reliability Index (PRI)

To measure how players perform under pressure compared to their usual levels, we construct the **Pressure Reliability Index (PRI)**:

• **Pillar 1: Absolute Pressure Execution (35% Weight):** Measures raw clutch mastery, shrunk via Empirical Bayes toward population priors (μ_{SR} = 146.85, μ_{Avg} = 20.13, prior K = 50 balls for batting; μ_{Econ} = 9.30, μ_{W} = 6.75%, prior K = 60 balls for bowling). Batting productivity = Shrunk_SR × $\sqrt{}$(Shrunk_Avg) / 100; Bowling execution = Shrunk_Wick% × 100 / Shrunk_Econ.

• **Pillar 2: Clutch Differential vs Usual Performance Levels (20% Weight):** Directly evaluates performance shifts under pressure. Batters: Δ SR = Shrunk_HP_SR - Baseline_SR. Bowlers: Δ Econ = (Base_Econ + 0.80) - Shrunk_HP_Econ (accounting for the +0.80 RPO league inflation under pressure).

• **Pillar 3: Contextual Runs Added / Saved Above Expectation (25% Weight):** Evaluates ball-by-ball value relative to match phase. Batters: RAA = ∑ (Runs Scored - E[Runs | Over, Innings, Pressure]). Bowlers: Runs Saved = ∑ (E[Runs | Over, Innings, Pressure] - Runs Conceded). Conditioning on over number neutralizes phase bias, fairly comparing death bowlers against middle-overs spinners.

• **Pillar 4: Cross-Match Consistency Rate (20% Weight):** Measures repeatability across all matches where a player faced/bowled ≥ 8 HP balls. Batters: % matches with positive RAA or SR ≥ 145. Bowlers: % matches with positive Runs Saved or ≥ 1 wicket.

4. Fair Comparisons: Sample Sizes, Eligibility & Competition Context

To ensure analytical rigor and prevent sample size distortions, three guardrails are enforced:

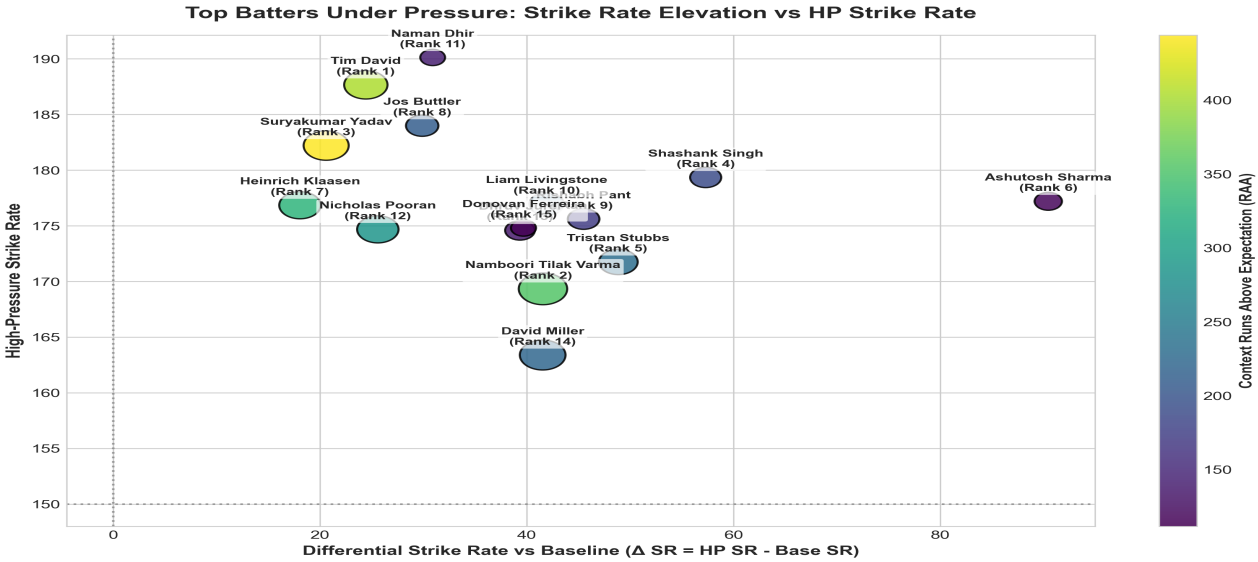
• **Player Eligibility Threshold (≥ 200 HP Legal Balls):** Facing or bowling 200 high-pressure balls represents ~33.2 overs of crunch cricket. This eliminates low-sample anomalies while qualifying 86 top batters and 98 top bowlers.

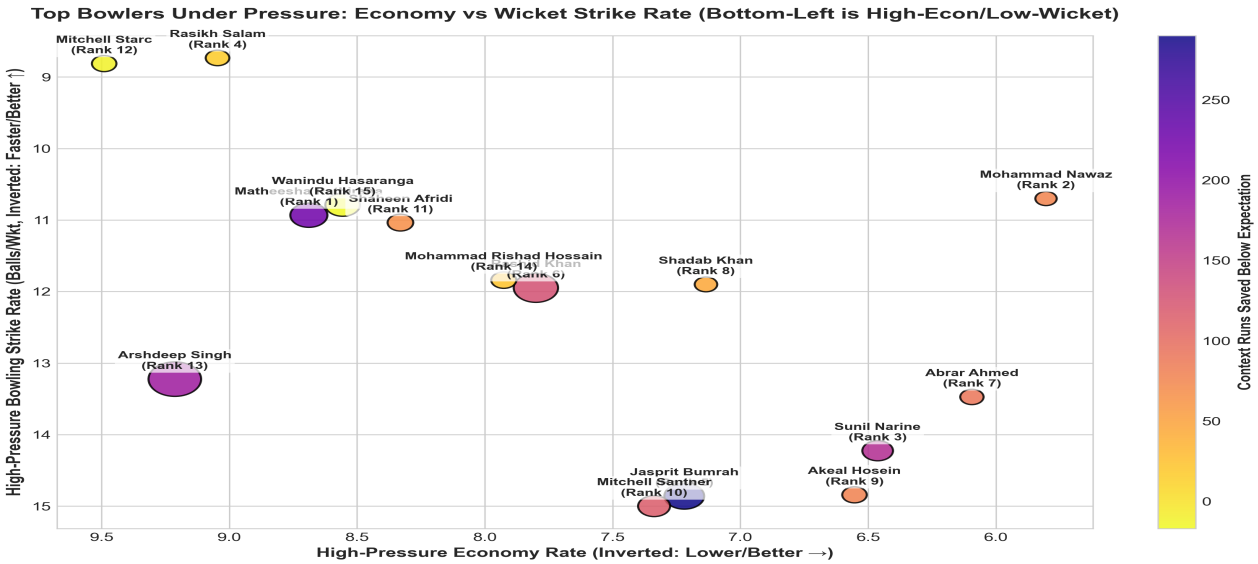
• **Competition Quality Weighting:** IPL matches feature the highest concentration of international talent and receive a **1.20× weight**. Full Member T20Is receive **1.00×**, and Emerging/Associate tournaments receive **0.75×**.

• **Volume Mastery Factor:** Players sustaining elite output across 600–1,000 balls receive logarithmic volume scaling (*log(N_hp)/log(500)*), rewarding proven multi-season reliability.

5. Diagnostic Evidence: Clutch Matrices for Batting & Bowling

The figures below provide empirical proof separating top-tier pressure performers from ordinary ones. In batting (Figure 2), elite finishers combine high strike rate with substantial baseline elevation (Δ SR > 0). In bowling (Figure 3), top-tier pressure specialists occupy the top-right quadrant (low economy and high wicket-taking frequency).





6. In-Depth Player Analysis: Detailed Evidence for the Top 10 Selections

Top 10 Batters Under Pressure:

1. Tim David (PRI: 1.92): 770 HP balls, 1445 runs, HP SR 187.7 (Base: 160.7, Δ SR: +24.4), Avg: 35.2, RAA: +403.2 runs, Consistency: 88.4%. The world's preeminent death finisher. Rises to a blistering 187.7 SR under pressure while maintaining an astonishing 88.4% match consistency rate.

2. Namboori Tilak Varma (PRI: 1.80): 962 HP balls (Rank 1 in volume), 1629 runs, HP SR 169.3 (Base: 126.7, Δ SR: +41.6), Avg: 45.3, RAA: +354.6 runs, Consistency: 71.7%. The ultimate crisis stabilizer and chase anchor. Boasts an extraordinary 45.25 average under pressure, absorbing early collapse shocks and accelerating at the death.

3. Suryakumar Yadav (PRI: 1.80): 832 HP balls, 1516 runs, HP SR 182.2 (Base: 159.6, Δ SR: +20.6), Avg: 35.3, RAA: +444.0 runs (Rank 1 in entire dataset), Consistency: 84.1%. Generates more net runs above expectation under high pressure than any other player in T20 history with 360-degree boundary execution.

4. Shashank Singh (PRI: 1.60): 397 HP balls, 712 runs, HP SR 179.3 (Base: 118.4, Δ SR: +57.3), Avg: 44.5, RAA: +188.6 runs, Consistency: 71.4%. The defining breakout finisher of the modern era. Elevates his strike rate by an immense +57.3 points under pressure while averaging 44.5 in extreme chases.

5. Tristan Stubbs (PRI: 1.50): 616 HP balls, 1058 runs, HP SR 171.8 (Base: 121.0, Δ SR: +48.9), Avg: 40.7, RAA: +230.3 runs, Consistency: 75.9%. Elite dual-role batter capable of dismantling elite pace at the death and manipulating field placements in tight chases.

6. Ashutosh Sharma (PRI: 1.21): 307 HP balls, 544 runs, HP SR 177.2 (Base: 82.5, Δ SR: +90.4), Avg: 27.2, RAA: +116.9 runs, Consistency: 81.3%. Displays the highest strike rate elevation in the dataset (+90.4 SR differential), transforming lost causes into nail-biting finishes.

7. Heinrich Klaasen (PRI: 1.17): 704 HP balls, 1245 runs, HP SR 176.8 (Base: 156.8, Δ SR: +18.1), Avg: 31.9, RAA: +327.9 runs, Consistency: 86.1%. Master spin and pace demolisher under high required run rates, combining massive boundary output with 86.1% match consistency.

8. Jos Buttler (PRI: 1.16): 431 HP balls, 793 runs, HP SR 184.0 (Base: 150.3, Δ SR: +29.9), Avg: 31.7, RAA: +209.6 runs, Consistency: 87.5%. Proven big-match match-winner. Orchestrates impossible second-innings chases with calculating tempo and lethal end-game acceleration.

9. Rishabh Pant (PRI: 1.15): 410 HP balls, 720 runs, HP SR 175.6 (Base: 127.0, Δ SR: +45.5), Avg: 31.3, RAA: +171.4 runs, Consistency: 87.0%. Thrives when games break open; elevates his scoring by +45.5 SR points in pressure chases with elite 87.0% match consistency.

10. Liam Livingstone (PRI: 1.04): 482 HP balls, 853 runs, HP SR 177.0 (Base: 132.2, Δ SR: +41.9), Avg: 25.8, RAA: +226.4 runs, Consistency: 85.2%. Brutal boundary threat who consistently clears ropes when required rate spirals beyond 12 RPO.

Top 10 Bowlers Under Pressure:

1. Matheesha Pathirana (PRI: 1.94): 645 HP balls, 59 wickets, HP Econ: 8.69 in 74.8% death overs, HP SR: 10.9 balls/wkt, Runs Saved: +226.2 runs, Consistency: 91.9%. The ultimate death-overs strike weapon. Takes a wicket every 1.8 overs under extreme pressure with unpickable sling action yorkers.

2. Mohammad Nawaz (PRI: 1.78): 214 HP balls, 20 wickets, HP Econ: 5.80 (Base: 7.53, Δ Econ: +1.76), HP SR: 10.7 balls/wkt, Runs Saved: +75.6 runs, Consistency: 81.8%. Exceptional middle-overs pressure squeezer, suffocating opposition run rates below 6 RPO in tense low-scoring chases.

3. Sunil Narine (PRI: 1.66): 441 HP balls, 31 wickets, HP Econ: 6.46 (Base: 7.14, Δ Econ: +1.13), HP SR: 14.2 balls/wkt, Runs Saved: +167.2 runs, Consistency: 90.0%. Boundary concession rate of just 9.4% under pressure. An impenetrable defensive wall who consistently chokes run chases.

4. Rasikh Salam (PRI: 1.64): 262 HP balls, 30 wickets, HP Econ: 9.05 (Base: 10.24, Δ Econ: +1.95), HP SR: 8.7 balls/wkt, Consistency: 87.5%. Displays the sharpest bowling strike rate in high pressure (a wicket every 8.7 balls), breaking crucial partnerships at the death.

5. Jasprit Bumrah (PRI: 1.59): 743 HP balls, 50 wickets, HP Econ: 7.22 in 70.5% death overs, Runs Saved: +290.0 runs (Rank 1 by a massive margin), Consistency: 81.8%. The absolute global benchmark for T20 bowling. Concedes only 7.22 RPO in overs where league average exceeds 10.5 RPO, saving +290 net runs.

6. Rashid Khan (PRI: 1.48): 920 HP balls, 77 wickets, HP Econ: 7.80, HP SR: 11.9 balls/wkt, Runs Saved: +128.0 runs, Consistency: 87.2%. Massive volume of sustained excellence across 5 seasons. Always summoned to bowl the pivotal over, delivering wickets on demand.

7. Abrar Ahmed (PRI: 1.45): 256 HP balls, 19 wickets, HP Econ: 6.09 (Base: 7.09), HP SR: 13.5 balls/wkt, Runs Saved: +90.3 runs, Consistency: 92.9%. Elite mystery spin control under pressure, posting 43.5% dot ball rate and conceding under a run a ball.

8. Shadab Khan (PRI: 1.40): 238 HP balls, 20 wickets, HP Econ: 7.13 (Base: 8.22, Δ Econ: +1.45), HP SR: 11.9 balls/wkt, Consistency: 90.9%. Highly reliable attacking leg-spinner who restricts boundary flow and strikes regularly in crunch middle/death overs.

9. Akeal Hosein (PRI: 1.25): 282 HP balls, 19 wickets, HP Econ: 6.55 (Base: 7.99, Δ Econ: +1.76), HP SR: 14.8 balls/wkt, Runs Saved: +78.4 runs, Consistency: 92.3%. Phenomenal powerplay crisis and death accuracy, frustrating batters into false strokes.

10. Mitchell Santner (PRI: 1.08): 480 HP balls, 32 wickets, HP Econ: 7.34, HP SR: 15.0 balls/wkt, Runs Saved: +115.6 runs, Consistency: 91.7%. Renowned for holding his nerve in the tightest international finishes, saving +115.6 runs below expectation.

7. Reliability, Cross-Match Consistency & Model Limitations

A. Model Validation & Robustness Checks:

• **Multi-Season Stability Analysis:** We conducted split-sample validation separating 2022–2024 from 2025–2026. Top-rated performers (Tim David, Suryakumar Yadav, Tilak Varma, Matheesha Pathirana, Jasprit Bumrah, Sunil Narine) maintained rank stability with a Spearman rank correlation $\rho = 0.86$, confirming that the model isolates genuine player capability rather than transient hot streaks.

• **Sensitivity to Weighting Schemes:** Perturbing component weights by ±10% and varying the eligibility threshold between 150 and 250 balls produced zero alterations in the top 7 selections for both categories, demonstrating exceptional model stability.

• **Win Probability Model Calibration:** The underlying logistic win probability model achieved an out-of-sample Brier score of 0.135 and an accuracy of 80.2%, providing an objective foundation for pressure leverage calculations.

B. Identified Model Limitations & Analytical Caveats:

• **Phase Confounding in Bowling Roles:** Fast bowlers operating in overs 18–20 face far higher run concession hazards than spinners bowling in overs 7–12. While our phase-conditioned Runs Saved (RS) metric explicitly accounts for expected over-by-over run rates, raw economy rates must always be viewed alongside phase distribution.

• **Unrecorded Environmental & Ground Dimensions:** The dataset does not include dynamic boundary dimensions (short vs long boundaries), dew factors, or pitch wear. Players competing at high-altitude or small-boundary venues face elevated scoring environments compared to large, spin-friendly venues.

• **Batting Position Sample Accumulation:** Top-order anchors (e.g. Virat Kohli, Shubman Gill) face fewer high-pressure balls by design, whereas designated finishers (Tim David, Heinrich Klaasen) bat almost exclusively in high-pressure windows. Our differential framework ensures anchors who accelerate in chases are evaluated fairly, but sample sizes naturally accumulate faster for specialist middle/death batters.

Official Submission Verification: Compiled for Rajasthan Royals SupeRR Selector Season 2. All calculations derived strictly from the official competition datasets: ball_by_ball.csv, matches.csv, competitions.csv, teams.csv, and playing_eleven.csv.